Covalent Nomenclature

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- Compound formed between two nonmetals (binary)
- Use the prefix system
  - Less electronegative atom comes first
  - Add prefixes to indicate the # of atoms. Omit *mono-* prefix on the first element
  - Add the ending *-ide* to the root of the second element

(Prefix+nonmetal #1) (Prefix+root of nonmetal #2-ide)

Prefixes

Prefix	Number
mono	1
di	2
tri	3
tetra	4
penta	5
hexa	6
hepta	7
octa	8
nona	9
deca	10

Nonmetal Roots

Hydrogen = Hydride	Boron = Bor-
Carbon = Carbide	Nitrogen = Nitr-
Oxygen = Oxide	Fluorine = Fluor-
Silicon = Silic-	Chlorine = Chlor-
Sulfur = Sulf-	Selenium = Selen-
Arsenic = Arsen-	Phosphorous = Phosph-
Bromine = Brom-	Iodine = Iod-

CO_2 carbon dioxide H_2O dihydrogen monoxide

Nomenclature: Formula → Name

CCl_4 carbon tetrachloride

N_2O dinitrogen monoxide

SF_6 sulfur hexafluoride

P_3Br_7 triphosphorus heptabromide

Si_5N_6 pentasilicon hexanitride

Nomenclature: Name → Formula

Arsenic trichloride $AsCl_3$

Dinitrogen pentafluoride N_2F_5

Tetraphosphorous decoxide P_4O_{10}

Hexacarbon dinitride C_6N_2

Boron Triiodide BI_3